सहयोगी प्राध्यापक (स्वयंचल अभियांत्रिकी)
महाराष्ट्र अभियांत्रिकी शिक्षक सेवा, गट-अ
उच्च व तंत्र शिक्षण विभाग
Associate Professor (Automobile Engineering)
Maharashtra Engineering Teachers' Service, Group-A
Higher and Technical Education Department

परीक्षेचे टप्पे: **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११२९)	इंग्रजी	१००	२००	एक तास	पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक या विषयांमध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ.क्र.	घटक व उपघटक
	AUTOMOBILE ENGINEERING
1	**Applied Thermodynamics:**
	Thermodynamic Concepts, First Law of Thermodynamics, One Dimensional Flow of Compressible Fluid, Second Law of Thermodynamics, Availability, Properties of Steam.
2	**Strength of Materials:**
	Simple Stress and Strain, S. F. and B. M. in Beams, Simple Theory of Bending, Sheer Stress in Beams, Simple Theory of Torsion, Bending Moment combined with Torsion and Axial Loads, Principal Stresses, Deflection of Beams, Strain Energy, Theories of failure.
3	**Material Science:**
	Strain Hardening, Constitution of Alloys, Iron-Carbon Equilibrium Diagram, Heat-Treatment of Steels, Cast Irons, Introduction to International Standards/Codes, Non- Ferrous Metals and Alloys, Fatigue Failure, Creep, Alloy Steels, Strengthening mechanism, Powder Metallurgy.
4	**Heat Transfer:**
	Conduction,Convection and Radiation, Emissivity, Heat exchangers, Mass Transfer (Mechanism, Fick's Law of Diffusion, Isothermal evaporation of water into air, Convective Mass Transfer)
5	**Machine Design and Vibration:**
	Design consideration in Castings and Forgings, Theories of Failure, Design for Static Loading, Design against Fluctuating Loads, Design of Shafts, Design of Springs, Design of Belts. Free un-damped single degree of freedom vibration system, Free damped single degree of freedom vibration system, Free un-damped multi degree of freedom vibration system, Forced single degree of freedom vibration system, Vibration measuring system, Rotor Dynamics, Balancing.
6	**Theory of Machines:**
	Basic Kinematics, Special Mechanisms, Velocity Analysis of Mechanisms, Acceleration Analysis of Mechanism, Static and Dynamic force analysis, Flexible Connector Mechanisms, Spur Gear Mechanism, Gear Trains, Cam Mechanism, Clutches, Brakes Dynamometer, Cam and followers, Governors, Gyroscope, (Static and Dynamic balancing of single rotor and multi rotor system. In-

	line Engines, Radial Engines, V-Engines.) Vibrations (Classification of Vibrations), Free undamped multi Degree of freedom vibratory systems, Forced single degree of freedom vibratory system.
7	**Internal Combustion Engines:**
	Classification of I.C. Engines, Cycle Analysis of I.C. Engines, S.I. Engines, C.I. Engines, Supercharging / Turbo charging, Performance Characteristics of S.I. and C.I. Engines, Air Pollution due to I.C. Engines, Fuels of I.C. Engines, Engine Lubrication, Engine Cooling, Alternative Fuels, Introduction to CNG, LPG, Stratified Charge and Wankel engines, Recent Developments in I.C. Engines.
8	**Vehicle Dynamics:**
	Fundamentals of Vehicle Dynamics, Road loads, Aerodynamic Drag, Lift, Rolling, Steering Geometry Parameters, Cornering Force, Cornering Power, Recent trends in Vehicle Dynamics, Vehicle Sensors, Central Tyre inflation systems, Sources of noise and vibration, pass-by noise requirements, target vehicles and objective targets, Vehicle structure noise, Engine noise, Transmission noise, Exhaust noise, causes and remedies on road shocks.
9	**Fluid Mechanics:**
	Fluid Kinematics, Fluid Dynamics, Real Fluid Flows, Boundary Layer Flows, Compressible Fluid Flow, Hydraulic Machinery (Turbines, Pumps, Compressors).
10	**Measurement and Metrology:**
	Static characteristics, Displacement measurement, Strain measurement, Measurement of angular velocity, Pressure measurement, Temperature measurement, Vacuum measurement, Acceleration measurement, Metrology.
11	**CAD/CAM:**
	Introduction and Elements of Interactive Computer Graphics, Techniques for Geometric Modeling, Transformation, Manipulation and Data Storage, NC and CNC Technology, Group Technology, CAPP and CAQC, Computer Integrated Manufacturing and Technology Driven Practices.
12	**Vehicle Systems, Chassis And Body Engineering:**
	Types of Clutches used in Automobiles, Braking systems, Steering system, Suspension system, Drive line, Final Drive, and Rear axels, Wheels and Tyres, Recent trends in Automobile Systems. Classification of motor Vehicles, Vehicle body materials, Visibility and Blind area, Driver's seat design requirements, Latest trends in Design, Manufacturing and Materials.
13	**Finite Element Analysis:**
	Introductory Concepts: Introduction to FEM, Brief History, General FEM Procedure, Applications of FEM in various fields, Advantages and Disadvantages of FEM Differential Equations in different Approximate solution of differential equations in fields, Matrix Algebra, FEM Procedure, Minimization of functional, Principle of minimum total potential, Piecewise Rayleigh-Ritz method, example problems in one-D, Isoperimetric, Algorithms for solution of equations dimensional structural analysis, Two dimensional finite element formulations, Finite element formulation of dynamics.
14	**Vehicle Testing And Maintenance:**
	Specification and Classification of Vehicles (including M, N and O layout), Regulations overview (ECE, EEC, FMVSS, AIS, CMVR, ADR), Homologation and its Types, Type approval and Conformity of Production, Types of World Harmonized Tests for Vehicles. Important Test tracks: Proving ground testing, high speed track, pavement track, corrugated track, mud track, steering pad, gradient track, Water/salt water wade track, Straight line braking track, split mu track, wet pad, Accelerated fatigue track, External noise test track, comfort track. Engine Tuning and vehicle care, Engine Diagnostics and testing, Engine overhaul, Diagnosis of various engine faults, Inspection, service and troubleshooting of ignition system, Starter motor and alternator, Latest diagnostic equipments, Wheel balancing and alignment, On boards diagnostics.
15	**Hybrid and Electric Vehicles**
	History and evolution of Electric Vehicles, Comparison of Electric with Internal Combustion Engine Vehicles, Exhaust Emission and Global warming, Environmental importance of Hybrid and Electric Vehicles, Overview of EV Challenges. Classification of EV, Overview of EV Technologies, Architecture, Construction, Working, Economic and Environmental impacts of using Electrical Vehicles, Emerging Technologies for Electric Vehicle Drives, Brief introduction to Autonomous and self- driving Vehicles. Classification of HEV, Architecture, Construction, Working, Tractive force, Power and Energy requirements for standard drive cycles of HEV, Power flow Control in Hybrid Drive-Train Topologies, Fuel Efficiency Analysis Energy Management Systems, Energy management strategies used in hybrid and electric vehicles,

	classification of different energy management strategies, comparison of different energy management strategies, implementation issues of energy management strategies, Battery Thermal Management Systems and its types, working and comparison.
16	**Automotive industry standards**
	Key industry quality standards, need, advantages, Cost of quality and value of quality, Vehicle safety norms, Indian Standards (IS) and Automotive Industry standards (AIS), Motor Vehicle Act (MVA) and Central motor vehicle Rule (CMVR), Bharat NCAP, Deming"s cycles and 14 Points, Juran Trilogy approach, Seven Quality Tools, Introduction to N Seven Tools, Quality Circle, 5S, Kaizen, Poka yoke, Kanban, JIT, ISO14001, Six Sigma, Criteria for Quality Award (National and International)

दिनांक – ३० एप्रिल, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग